Plots — the rest of the toolbox

TC1037 · Algorithmic Thinking · Session 14b · complementary notes

Dr. Leopoldo Rodríguez Salazar

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Tecnológico de Monterrey · Campus Querétaro · Group 510

What this is

You have seen three charts: the bar, the line, and two of them side by side. That is everything the lab and the project strictly require.

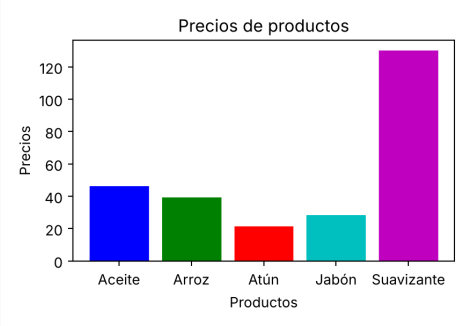
This deck is the rest of the toolbox — four more chart types, the format strings, the legend, and the five ways a plot goes wrong.

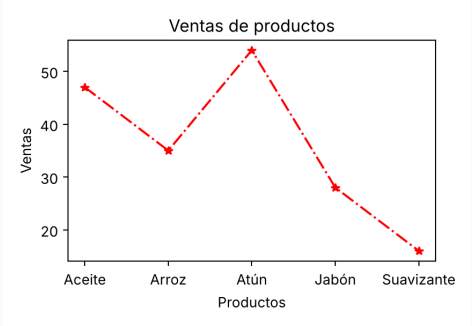
Read it when you need it.

Nothing here is examined beyond what was in Tuesday's session. It is here so that when your question does not fit a bar chart, you are not stuck.

Everything in it is one line of `matplotlib`.

Recap — the two you already have






```
plt.bar(nombres, valores)
```

```
plt.plot(x, y, "r*-.")
```

Compares separate things. Categories have no order — you may sort them however you like.

Follows one thing changing. The x axis must have an order — months, years, steps. Products do not.

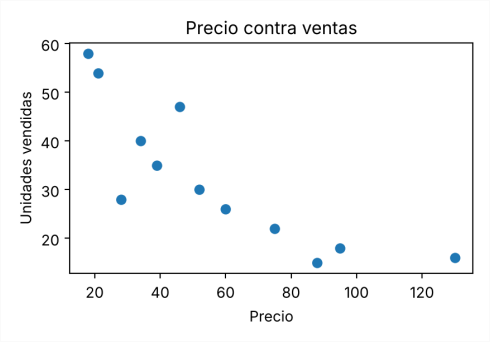
Both need `plt.title`, `plt.xlabel`, `plt.ylabel`. Always.

Scatter — when both columns are numbers

```
plt.title("Precio contra ventas")
plt.xlabel("Precio")
plt.ylabel("Unidades vendidas")
plt.scatter(precio, venta)
plt.show()
```

Use it when your question is *does one thing move with another?*

No grouping needed — one dot per row, so this is the one chart you can draw straight from your forty rows.



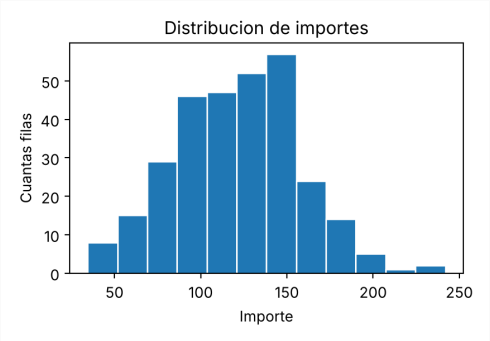
"Do the expensive products sell less?" — the dots answer it before you compute anything.

Histogram — the shape of one column

```
plt.title("Distribucion de
    importes")
plt.xlabel("Importe")
plt.ylabel("Cuantas filas")
plt.hist(importes, bins=12)
plt.show()
```

One numeric column, no second list. It counts how many rows fall in each band.

Answers *"are most of our orders small, or is it a few big ones?"* — which a mean alone hides completely.



`bins` is how many bars. Try 10 and try 30 — the story can change.

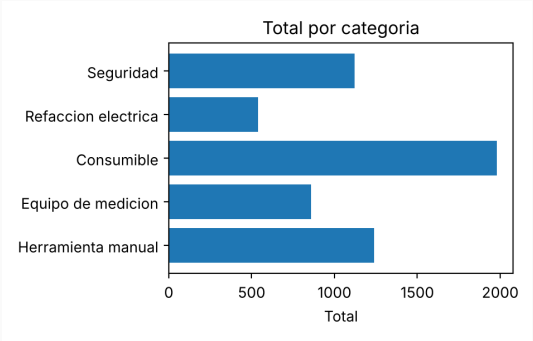
barh — when the labels are long

```
plt.barh(categorias, totales)
```

Exactly `bar`, turned on its side.

Use it the moment your category names overlap each other on the x axis — which happens with anything longer than about eight characters.

Careful: `xlabel` is now the *value* and `ylabel` the category. Swapping them is the usual slip.



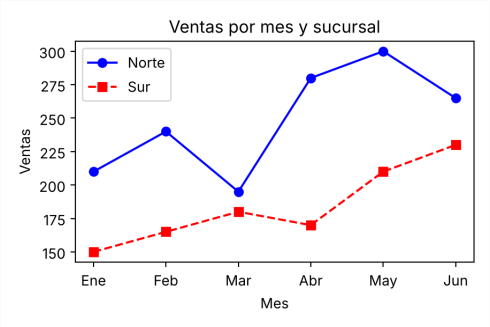
Two series on one chart — and the legend

```
plt.plot(meses, norte, "bo-",
    label="Norte")
plt.plot(meses, sur,   "rs--",
    label="Sur")
plt.legend()
plt.show()
```

Call `plt.plot` twice before `show()`. Both lines land on the same axes.

The moment there is more than one series, the legend stops being optional — it is in the project rubric.

`label=` on each line, then `plt.legend()` once.



This is one of the three shapes `grafica2` is allowed to take: one x against two y's.

The format string, in full

Colour	Marker	Line
`b` blue `g` green	`o` circle `s` square	`-` solid
`r` red `c` cyan	`*` star `+` plus	`--` dashed
`m` magenta `k` black	`.` dot `x` cross	`-.` dash-dot `:` dotted

```
plt.plot(x, y, "r*-.")
plt.plot(x, y, "go--")
plt.plot(x, y, "k.")
```

Any order, any subset. `"k."` is black dots with no line at all — a poor man's scatter.

For bars it is different: `plt.bar` hands you back the bars, and you colour them one by one with `barras[0].set_color('b')`.

Which chart — decide from the question, not from taste

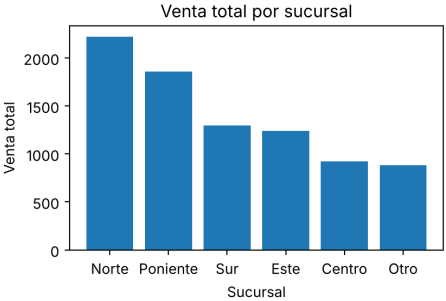
Your question sounds like	You need	Chart
Which category is biggest?	group, then total or mean	`bar` (or `barh` if the names are long)
Did it grow over time?	an ordered x axis	`plot`
Do two numbers move together?	nothing — raw rows	`scatter`
Are most values small or large?	one numeric column	`hist`
Two groups compared over time?	two lists of y	two `plot` calls + `legend`
Two different questions at once?	both of the above	`subplot(121)` and `(122)`

Rows one, two and five are what almost every project needs. The rest are there when your data does not cooperate.

The pie chart, and why I would rather you did not



Rank Sur against Este. Go on.



The same six numbers. You ranked them without trying.

`plt.pie(valores, labels=nombres)` works, and it is not forbidden. But people read **length** far better than **angle**, so a pie hides exactly the comparison you drew it to show. Use one only for *"roughly half"*, and never with more than four slices.

Five ways a plot goes wrong

What you see	What happened
Two windows, the second one empty	`plt.show()` called *between* the two `subplot` blocks. One `show()`, at the end.
`ValueError: x and y must be the same size`	Your two lists are different lengths. You appended to one inside an `if` and to the other outside it.
Forty unreadable bars	You plotted the rows instead of grouping first.
Everything piled on one chart	You never closed the previous figure. `plt.show()` clears it; without it, the next plot lands on top.
A beautiful chart that says nothing	No title, no axis labels. The marker is not allowed to guess, and neither is your reader.

What the project actually asks for

`grafica1`	**A simple bar or line chart.** One question, answered.
`grafica2`	**One of three**: two sub-plots in one figure · one x against two y's · a `groupby` chart.
Both	**Title, axis labels, and a legend when there is more than one series.** Every time.

Everything else in this deck is optional. It exists so that when your question does not fit a bar chart, you have somewhere to go instead of changing the question.